HARE.jl — Implementation Notes

This document describes, for each estimator in HARE.jl, the exact computational procedure, the choices made at each step, and the references supporting those choices. It is intended as an internal reference for contributors and for users who want to understand what the code is actually doing.

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1. Shared infrastructure

1.1 Rho estimation (Cochrane-Orcutt moment estimator)

$$\hat{\rho} = \frac{\sum_{t=2}^{n} u_t \, u_{t-1}}{\sum_{t=2}^{n} u_{t-1}^2}$$

This is the standard first-order autoregressive moment estimator proposed by Cochrane & Orcutt (1949). It is consistent under mild regularity conditions and cheap to compute. It is used as the initial estimate of rho in all two-step AR(1) procedures.

1.2 Prais-Winsten transformation

For $\rho \in (-1, 1)$, the AR(1) error structure is removed by quasi-differencing:

$$\begin{aligned} y_1^* &= \sqrt{1-\rho^2}\, y_1 \\ y_t^* &= y_t - \rho\, y_{t-1}, \quad t = 2, \dots, n \\ \mathbf{X}_{1,:}^* &= \sqrt{1-\rho^2}\, \mathbf{X}_{1,:} \\ \mathbf{X}_{t,:}^* &= \mathbf{X}_{t,:} - \rho\, \mathbf{X}_{t-1,:}, \quad t = 2, \dots, n \end{aligned}$$

The first-observation scaling by $\sqrt{1-\rho^2}$ is the Prais-Winsten correction (Prais & Winsten, 1954). The alternative — simply dropping the first observation — is the Cochrane-Orcutt convention. Prais-Winsten is preferred because it retains all n observations and is asymptotically more efficient, particularly when ρ is large and n is small.

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2. Harvey (1976) FWLS — multiplicative heteroskedasticity

Model

$$y_i = \mathbf{x}_i^\top \beta + u_i, \quad u_i \sim \mathcal{N}(0, \sigma_i^2), \qquad \log \sigma_i^2 = \mathbf{z}_i^\top \gamma$$

where $\mathbf{Z}$ is an auxiliary regressor matrix (by default equal to $\mathbf{X}$ augmented with an intercept; user-supplied $\mathbf{Z}$ must not include a constant — one is prepended internally, consistent with the $\mathbf{X}$ convention).

Procedure

Step 1. OLS to get residuals $\hat{u}$.

Step 2. Auxiliary log-linear regression for γ. Because $u_i = \sigma_i \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0,1)$:

$$\log u_i^2 = \log \sigma_i^2 + \log \varepsilon_i^2 = \mathbf{z}_i^\top \gamma + \log \varepsilon_i^2$$

$\log \varepsilon_i^2$ is distributed as $\log \chi^2(1)$, which has mean

$$\mathrm{E}[\log \chi^2(1)] = \psi(1/2) + \log 2 = -1.2703628454614782 =: c$$

The naive OLS of $\log \hat{u}^2$ on $\mathbf{Z}$ therefore estimates γ with all coefficients biased by $(\mathbf{Z}^\top\mathbf{Z})^{-1}\mathbf{Z}^\top c\mathbf{1}$, where $c = \mathrm{E}[\log \chi^2(1)]$. When regressors have non-zero means this contaminates every element of γ, not just the intercept.

The correction subtracts c from the dependent variable **before** fitting:

$$\log \hat{u}_i^2 - c = \mathbf{z}_i^\top \gamma + v_i, \qquad \mathrm{E}[v_i] = 0$$

Because $c\mathbf{1}$ lies in the column space of $\mathbf{Z}$ (an intercept column is always present), OLS absorbs the shift exactly into the intercept, yielding consistent estimates of **all** elements of γ regardless of regressor means. The fitted values $\mathbf{z}_i^\top \hat{\gamma}$ then directly estimate $\log \sigma_i^2$.

Step 3. WLS with weights $w_i = 1/\exp(\mathbf{z}_i^\top \hat{\gamma})$.

Step 4. Recover β by WLS of $\mathbf{y}$ on $\mathbf{X}$ with weights $\mathbf{w}$.

Iterated version

Repeat Steps 2–4 until $\max|\beta^{(i)} - \beta^{(i-1)}| < \text{tol}$. Convergence of the iterated FWLS to the MLE is established under regularity conditions by Oberhofer & Kmenta (1974).

`exponential_ifwls` is the preferred FWLS choice when the iterative reweighting framework is explicitly desired. For asymptotically efficient estimation, `exponential_mle` (Section 4) achieves the MLE directly via joint L-BFGS.

Standard errors

Reported vcov is the WLS sandwich $\hat{\sigma}^2(\mathbf{X}^\top\mathbf{W}\mathbf{X})^{-1}$, conditional on the estimated weights. This is the standard feasible GLS standard error; it does not account for the estimation uncertainty in γ.

References

Harvey, A. C. (1976). Estimating regression models with multiplicative heteroscedasticity. *Econometrica*, 44(3), 461–465.

Oberhofer, W., & Kmenta, J. (1974). A general procedure for obtaining maximum likelihood estimates in generalized regression models. *Econometrica*, 42(3), 579–590.

Greene, W. H. (2018). *Econometric Analysis* (8th ed.). Pearson, §9.7.

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3. Glejser (1969) FWLS — linear standard deviation model

Model

$$y_i = \mathbf{x}_i^\top \beta + u_i, \quad u_i \sim \mathcal{N}(0, \sigma_i^2)$$
$$\sigma_i = \gamma_0 + \mathbf{z}_i^\top \gamma$$

where $\mathbf{Z}$ is an auxiliary regressor matrix (by default equal to $\mathbf{X}$ without the intercept column; a constant is prepended internally, consistent with the $\mathbf{X}$ convention).

Procedure

Step 1. OLS to get residuals $\hat{u}$.

Step 2. Auxiliary regression for γ. Under the model, $|u_i| = \sigma_i|\varepsilon_i|$ with $\varepsilon_i \sim \mathcal{N}(0,1)$, so

$$\mathrm{E}[|u_i|] = \sigma_i \,\mathrm{E}[|\varepsilon_i|] = \sigma_i\sqrt{2/\pi}$$

OLS of $|\hat{u}|$ on $\mathbf{Z}$ therefore consistently estimates $\gamma\sqrt{2/\pi}$, not γ itself. The correction multiplies the raw OLS coefficients by $\sqrt{\pi/2}$:

$$\hat{\gamma} = \sqrt{\pi/2}\,(\mathbf{Z}^\top\mathbf{Z})^{-1}\mathbf{Z}^\top|\hat{\mathbf{u}}|$$

The vcov is scaled accordingly by $\pi/2$. For the WLS weights, $\hat{\sigma}_i = \mathbf{z}_i^\top\hat{\gamma}$ and $w_i = 1/\hat{\sigma}_i^2$. The estimator stops with an error if any $\hat{\sigma}_i$ is zero or negative (below 10^{-6} of the mean fitted value), and the linear link does the same for $\hat{\sigma}_i^2$: no weight exists for such a row, and a floor would give it a weight that dominates the fit.

Without this correction the reported γ would be biased toward zero by the factor $\sqrt{2/\pi} \approx 0.798$ for all n, making the estimator appear inconsistent. The WLS β itself is unaffected because a constant weight scaling cancels in the normal equations, but the displayed γ values and their standard errors would be wrong.

Step 3. WLS of $\mathbf{y}$ on $\mathbf{X}$ with weights $\mathbf{w}$.

Iterated version

Repeat Steps 2–3 until $\max|\beta^{(i)} - \beta^{(i-1)}| < \text{tol}$.

`quadratic_ifwls` is the preferred FWLS choice when the iterative reweighting framework is explicitly desired. For asymptotically efficient estimation, `quadratic_mle` (Section 4) achieves the MLE directly via joint L-BFGS.

Standard errors

Same as Harvey: WLS sandwich conditional on the estimated weights.

References

Glejser, H. (1969). A new test for heteroskedasticity. *Journal of the American Statistical Association*, 64(325), 316–323.

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4. MLE heteroskedasticity estimators — exponential, quadratic, linear, power

Four estimators share a common MLE framework, differing only in the link function that maps the variance equation parameters to σ_i^2. Three correspond to the SAS `PROC AUTOREG HETERO` link options; the power link is HARE-native.

Models

Function	Link	Variance	Origin	Associated test
`exponential_mle`	EXP	$\sigma_i^2 = \exp(\gamma_0 + \mathbf{z}_i^\top\gamma)$	Harvey (1976)	Harvey LM test
`quadratic_mle`	SQUARE	$\sigma_i = \gamma_0 + \mathbf{z}_i^\top\gamma$	Glejser (1969)	Glejser test
`linear_mle`	LINEAR	$\sigma_i^2 = \gamma_0 + \mathbf{z}_i^\top\gamma$	Breusch & Pagan (1979)	Breusch-Pagan test (`HypothesisTests.BreuschPaganTe`
`power_mle`	POWER	$\log\sigma_i^2 = \gamma_0 + \sum_j \gamma_j \log\lvert z_{ij}\rvert$, $\gamma_j = 2\delta_j$	—	—

In all cases $\mathbf{z}_i$ is the user-supplied auxiliary regressor vector **without** a constant; γ_0 is the intercept added internally. The default is $\mathbf{z}_i = \tilde{\mathbf{x}}_i$ (the mean regressors without their intercept column).

Log-likelihood

All three maximise the normal log-likelihood jointly over $\theta = (\beta, \gamma_0, \gamma)$:

$$\ell(\theta) = -\frac{n}{2}\log(2\pi) - \frac{1}{2}\sum_i \log \sigma_i^2(\theta) - \frac{1}{2}\sum_i \frac{(y_i - \mathbf{x}_i^\top \beta)^2}{\sigma_i^2(\theta)}$$

For the exponential link the log-variance is linear in the parameters so the objective is globally concave in γ for fixed β. For the quadratic and linear links the objective can be non-concave; the warm start (see below) is important for reliable convergence.

Numerical stability

The quadratic and linear link functions require $\sigma_i > 0$ and $\sigma_i^2 > 0$ respectively. The two links are handled differently.

Exponential link. $\sigma_i^2 = \exp(\mathbf{z}_i^\top \gamma)$ is strictly positive for any γ, so no clamping or smoothing is needed. The Harvey FWLS warm start also carries slope information (the auxiliary log-linear regression directly estimates each element of γ), so L-BFGS starts in a well-identified region and never explores infeasible directions.

Quadratic and linear links. Nothing keeps $\mathbf{z}_i^\top \gamma$ positive, so the negative log-likelihood returns `Inf` for any γ with a fitted value at or below zero. The line search steps back from that value, so the search stays inside the model. All of γ is free: γ_0 may be negative as long as every fitted value is positive, as it is at the maximum on some data. `crch` finds the same value there.

Why not a floor. A floor such as $\max(\sigma_i^2, 10^{-10})$ lets the search leave the model. A row that falls below the floor while the regression line passes through it adds up to $-\frac{1}{2}\log 10^{-10} \approx 11.5$ to the log-likelihood, so the search is drawn there and reports convergence at a spike. A floor also depends on the units of y. The likelihood itself is still unbounded as a variance approaches zero with a zero residual, so the estimators check the fit at the optimum and stop with an error if a fitted value has reached zero.

Why not softplus. Softplus ($\tilde{\sigma}^2 = \log(1 + e^x)$) is smooth and strictly positive everywhere, which makes it attractive in principle. However, it changes the model: the estimated γ no longer corresponds to the linear variance function $\sigma_i^2 = \mathbf{z}_i^\top \gamma$.

Optimisation

Minimisation of the negative log-likelihood using **L-BFGS** (Optim.jl) with gradients supplied by **ForwardDiff.jl** automatic differentiation. The full parameter vector $\theta = (\beta, \gamma_0, \gamma)$ is optimised jointly; the mean and variance equations are not concentrated.

Choice of L-BFGS over concentrated likelihood. Unlike Beach-MacKinnon or Joint HARE — where concentrating out β analytically reduces the problem dimension substantially — here the mean and variance parameters enter the gradient in a coupled way that does not yield a cheap closed-form profile. Joint L-BFGS with AD is straightforward and fast for the parameter dimensions typical of these models.

Choice of backtracking for the quadratic and linear links. HagerZhang, the default line search, fails an internal assertion on the infinite value outside the model; backtracking steps back from it. The cost is more iterations, for example 16 against 4 on one tutorial fit.

Units of y. The two links fit $y/\mathrm{sd}(y)$ and convert the results back, because their γ has the units of y or y^2, far from those of β when y is very small or very large.

Warm starts

Good starting values are essential for the quadratic and linear links.

Link	Warm start
Exponential	Two-step FWLS: β_0 from OLS, γ_0 from auxiliary log-linear regression of $\log \hat{u}^2 - c$ on Z
Quadratic	Two-step FWLS: β_0 from OLS, γ_0 from auxiliary regression of $\lvert\hat{u}\rvert$ on Z (with $\sqrt{\pi/2}$ correction)
Power	Two-step FWLS on $\log \lvert Z\rvert$: β_0 from OLS, γ_0 from auxiliary log-linear regression of $\log \hat{u}^2 - c$ on $[\mathbf{1}, \log \lvert Z\rvert]$
Linear	Two-step FWLS: β_0 from OLS, γ_0 from auxiliary regression of $\hat{u}^2$ on Z

If the two-step line of the quadratic or linear link has a fitted value at or below zero, the search starts from a constant instead: $\gamma_0 = \sqrt{\pi/2}\,\overline{|\hat{u}|}$ or $\gamma_0 = \overline{\hat{u}^2}$, with the other elements zero. The constant is not the first choice, because it can overshoot into the spike on data with a proper maximum nearby.

Standard errors

The full observed information matrix is computed by `ForwardDiff.hessian` of the negative log-likelihood at the optimum $\hat{\theta}$. The vcov of the complete parameter vector is H^{-1}. The β and γ blocks are then extracted:

$$\widehat{\mathrm{Var}}(\hat{\beta}) = H^{-1}[1{:}k,\ 1{:}k], \qquad \widehat{\mathrm{Var}}(\hat{\gamma}) = H^{-1}[k{+}1{:}\mathrm{end},\ k{+}1{:}\mathrm{end}]$$

This is the full-information MLE standard error, accounting for joint estimation of mean and variance parameters. It is asymptotically efficient under correct specification and normality, in contrast to the FWLS standard errors in Sections 2–3 which are conditional on the estimated weights.

Hessian robustness for the linear link

Problem. The curvature of the linear link's likelihood grows without limit as a fitted variance approaches zero, so when the optimum sits near that boundary the Hessian can fail to be positive definite (PD). On some platforms — in particular macOS ARM64 (Apple Silicon) running Julia 1.10 — floating-point evaluation order differs enough from x86 that the Hessian is numerically non-PD more frequently. `inv(H)` with a non-PD matrix produces negative diagonal entries in the vcov, making all reported standard errors imaginary and causing tests like `all(diag(m.gamma_vcov) .> 0)` to fail.

Alternatives considered.

Option	Why rejected
Softplus link	Changes the model semantics; destabilises convergence (see above)
Perturb Hessian: $H + \lambda I$	Arbitrary λ distorts standard errors; no principled choice
Cholesky with fallback	More invasive; still requires choosing a perturbation magnitude
Ignore non-PD (return NaN SEs)	Silently wrong output; fails CI
Finite-difference Hessian	Slower and less accurate than AD; does not fix the non-smoothness

Solution. After computing `H = ForwardDiff.hessian(nll, theta_hat)`:

1. Wrap as `Symmetric(H)` before inversion. This enforces exact symmetry (ForwardDiff can produce tiny asymmetries from floating-point cancellation) and allows `inv` to use a symmetric-specific LAPACK path.

2. If the result still has any non-positive diagonal entry (which only happens for the linear link on pathological platforms/seeds), fall back to `pinv(Symmetric(H))` — the Moore-Penrose pseudoinverse. `pinv` uses an SVD, thresholds near-zero singular values to zero, and thereby produces a PD (or at least positive-semidefinite) result even when `H` is numerically singular.

The fallback is guarded by `link === :linear` so the exponential, quadratic, and power links — which never produce non-PD Hessians in practice — are unaffected and pay no extra cost.

References

Harvey, A. C. (1976). Estimating regression models with multiplicative heteroscedasticity. *Econometrica*, 44(3), 461–465.

Glejser, H. (1969). A new test for heteroskedasticity. *Journal of the American Statistical Association*, 64(325), 316–323.

SAS Institute Inc. (2015). *SAS/ETS 14.1 User's Guide: The AUTOREG Procedure*. SAS Institute Inc., Cary, NC.

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5. Groupwise heteroscedasticity

Model

$$y_i = \mathbf{x}_i^\top \beta + u_i, \qquad \mathrm{Var}(u_i) = \sigma_g^2 \ \text{ for all } i \in g$$

Observations are divided into G groups. The variance is constant within each group but unrestricted across groups. Group membership is supplied by the user as a vector of labels (any type: integers, strings, …).

Procedure

Step 1. OLS to get residuals $\hat{u}_i$.

Step 2. Estimate group variances: $\hat{\sigma}_g^2 = \dfrac{1}{n_g} \sum_{i \in g} \hat{u}_i^2$.

Step 3. WLS with weights $w_i = 1/\hat{\sigma}_g^2$.

Iterated version and MLE equivalence

The MLE first-order condition for σ_g^2 — obtained by differentiating the normal log-likelihood — is exactly $\hat{\sigma}_g^2 = (1/n_g) \sum_{i \in g} \hat{u}_i^2$. Therefore the iterated FWLS (repeating Steps 2–3 until β converges) is the EM algorithm for this model and converges to the joint MLE of $(\beta, \sigma_1^2, \ldots, \sigma_G^2)$. `groupwise_mle` runs this iteration internally and is the recommended call when the MLE is the goal; `iterated_groupwise` is an alias that also converges to the MLE but is named for the iterative FWLS framework.

Standard errors

WLS vcov conditional on $\hat{\sigma}_g^2$, identical in form to the Harvey FWLS standard errors.

References

Greene, W. H. (2018). *Econometric Analysis* (8th ed.). Pearson, §9.7.2.

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6. Prais-Winsten / Cochrane-Orcutt — AR(1) autocorrelation

Model

$$y_t = \mathbf{x}_t^\top \beta + u_t, \qquad u_t = \rho\, u_{t-1} + e_t, \quad e_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$$

Two-step procedure

Step 1. OLS; estimate ρ by the Cochrane-Orcutt moment estimator (Section 1.1).

Step 2. Apply Prais-Winsten transformation (Section 1.2); OLS on the transformed system.

Iterated procedure

Alternate between: - re-estimating ρ from the residuals $\mathbf{y} - \mathbf{X}\hat{\beta}$, - re-transforming and re-fitting by OLS, until $|\rho^{(i)} - \rho^{(i-1)}| < \text{tol}$.

Choice: Prais-Winsten over Cochrane-Orcutt

Hildreth-Lu and Beach-MacKinnon use the Cochrane-Orcutt convention (dropping the first observation) because their objectives are defined that way in their respective papers. The Prais-Winsten estimator retains the first observation, which is asymptotically more efficient (the gain is $O(1/n)$ but can matter in small samples).

Standard errors

OLS vcov on the transformed system, conditional on $\hat{\rho}$.

References

Prais, S. J., & Winsten, C. B. (1954). Trend estimators and serial correlation. *Cowles Commission Discussion Paper*, No. 383.

Cochrane, D., & Orcutt, G. H. (1949). Application of least squares regression to relationships containing auto-correlated error terms. *Journal of the American Statistical Association*, 44(245), 32–61.

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7. Hildreth-Lu (1960) — grid search over rho

Procedure

For each ρ in a uniform grid of `nrho = 200` points on $(-0.99, 0.99)$:

1. Apply the Cochrane-Orcutt quasi-difference (first observation dropped): $y_t^* = y_t - \rho\, y_{t-1}$, $\mathbf{X}_t^* = \mathbf{X}_t - \rho\, \mathbf{X}_{t-1}$, $t = 2, \ldots, n$.
2. OLS on the transformed system; record RSS.

Select the ρ that minimises RSS.

Choices

- **Grid vs optimisation.** A grid search is guaranteed to find the global minimum over the search space and is robust when the RSS surface has multiple local minima. The cost is resolution: with 200 points the grid spacing is ≈ 0.01 in ρ. If higher precision is needed, `nrho` can be increased.
- **First observation dropped.** This follows the original Hildreth-Lu convention and is consistent with their derivation. The effective sample size is therefore $n - 1$.
- **Search interval** $(-0.99, 0.99)$**.** Values at ± 1 make the transformation degenerate; ± 0.99 avoids numerical issues while covering the practically relevant range.

Standard errors

OLS vcov on the transformed system at the selected ρ, conditional on $\hat{\rho}$.

References

Hildreth, C., & Lu, J. Y. (1960). Demand relations with autocorrelated disturbances. *Michigan State University Agricultural Experiment Station Technical Bulletin*, No. 276.

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8. Beach-MacKinnon (1978) — exact MLE for AR(1)

Log-likelihood

The exact log-likelihood for the AR(1) model (including the Jacobian for the first observation) is, up to constants:

$$L(\rho) = \frac{n}{2}\log(1-\rho^2) - \frac{n}{2}\log\left(\frac{S(\rho)}{n}\right)$$

where $S(\rho) = \|\mathbf{y}^* - \mathbf{X}^*\hat{\beta}(\rho)\|^2$ is the residual sum of squares at the GLS estimate $\hat{\beta}(\rho)$ obtained after the Prais-Winsten transformation at ρ. Concentrating out β and σ^2 analytically leaves a univariate problem in ρ.

The term $\frac{1}{2}\log(1-\rho^2)$ is the Jacobian contributed by the first observation under the stationary AR(1) distribution. Dropping it (as in the Prais-Winsten/Cochrane-Orcutt objective) gives an approximation that is $O(1/n)$ away from the exact likelihood.

Maximisation

Implemented as minimisation of the negative concentrated log-likelihood via **Brent's method** on the interval $(-0.999, 0.999)$. Brent's method is chosen because the problem is univariate, the function is smooth but not necessarily unimodal, and Brent's method combines the robustness of bisection with the speed of inverse quadratic interpolation.

Recovery of beta and sigma^2

Once $\hat{\rho}$ is found, $\hat{\beta}$ and $\hat{\sigma}^2$ are recovered by OLS on the Prais-Winsten-transformed system at $\hat{\rho}$.

Standard errors

Conditional on $\hat{\rho}$: the OLS vcov on the transformed system.

References

Beach, C. M., & MacKinnon, J. G. (1978). A maximum likelihood procedure for regression with autocorrelated errors. *Econometrica*, 46(1), 51–58.

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9. Sequential HARE — AR(1) + Harvey heteroskedasticity

Model

$$\begin{aligned} y_t &= \mathbf{x}_t^\top \beta + u_t \\ u_t &= \rho\, u_{t-1} + \sigma_t\, e_t, \quad e_t \overset{\text{iid}}{\sim} \mathcal{N}(0,1) \\ \log\sigma_t^2 &= \mathbf{x}_t^\top \gamma \end{aligned}$$

Two-step procedure

Step 1. OLS; estimate ρ by the Cochrane-Orcutt moment estimator.

Step 2. Prais-Winsten transformation $\rightarrow$ residuals $\widehat{u}^{\mathrm{pw}}$ from OLS on the transformed system.

Step 3. Harvey auxiliary regression of $\log(\widehat{u}_t^{\mathrm{pw}})^2 - c$ on the **original X** (not the transformed $\mathbf{X}^*$). Rationale: the PW innovations satisfy $\widehat{u}_t^{\mathrm{pw}} \approx \sigma_t\, e_t$ with $\log\sigma_t^2 = \mathbf{x}_t^\top\gamma$ in the original space, so the correct auxiliary regressors are the untransformed $\mathbf{x}_t$.

Step 4. WLS on the PW-transformed system $(\mathbf{X}^*, \mathbf{y}^*)$ with Harvey weights $w_t = 1/\exp(\mathbf{x}_t^\top\widehat{\gamma})$.

Iterated version

Alternate between Steps 1–4 until $\max\bigl(\max|\beta^{(i)} - \beta^{(i-1)}|,\ |\rho^{(i)} - \rho^{(i-1)}|\bigr) < \mathrm{tol}$.

References

Oberhofer, W., & Kmenta, J. (1974). A general procedure for obtaining maximum likelihood estimates in generalized regression models. *Econometrica*, 42(3), 579–590.

Harvey, A. C. (1976). Estimating regression models with multiplicative heteroscedasticity. *Econometrica*, 44(3), 461–465.

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10. Joint HARE — simultaneous MLE for AR(1) + Harvey heteroskedasticity

Model

Same as the Sequential HARE model (Section 9).

Log-likelihood

The exact log-likelihood (up to the constant $-\frac{n}{2}\log(2\pi)$) is:

$$\ell(\rho,\gamma,\beta) = \tfrac{1}{2}\log(1-\rho^2) - \tfrac{1}{2}\sum_t \log\sigma_t^2 - \tfrac{1}{2}\left(\frac{\sqrt{1-\rho^2}\,u_1}{\sigma_1}\right)^2 - \tfrac{1}{2}\sum_{t=2}^{n}\left(\frac{u_t - \rho\,u_{t-1}}{\sigma_t}\right)^2$$

where $u_t = y_t - \mathbf{x}_t^\top\beta$ and $\sigma_t = \sqrt{\exp(\mathbf{z}_t^\top\gamma)}$.

Two-step procedure

Step 1 (concentrated MLE). Profile out β analytically at each function evaluation: given (ρ,γ), apply the doubly-transformed system

$$y_t^{**} = y_t^*/\sigma_t, \qquad \mathbf{X}_t^{**} = \mathbf{X}_t^*/\sigma_t$$

where $\mathbf{y}^*$, $\mathbf{X}^*$ is the Prais-Winsten transformation at ρ, and compute $\widehat{\beta}(\rho,\gamma) = (\mathbf{X}^{**\top}\mathbf{X}^{**})^{-1}\mathbf{X}^{**\top}\mathbf{y}^{**}$ by OLS. The objective is then a function of (ρ,γ) only.

Step 2. Minimise the negative concentrated log-likelihood over $(\mathrm{atanh}(\rho),\gamma)$ using **L-BFGS** (Optim.jl). The unconstrained reparametrisation $\rho = \tanh(\xi)$ enforces $|\rho| < 1$ without box constraints. Gradients are computed by **ForwardDiff.jl** automatic differentiation. Once convergence is reached, β is recovered by one OLS step on the doubly-transformed system.

Iterated procedure

Coordinate descent:

1. With β fixed, minimise the negative log-likelihood over (ρ, γ) via L-BFGS.
2. With (ρ, γ) fixed, update β by one OLS step on the doubly-transformed system.

Repeat until $\max\big(\max|\beta^{(i)} - \beta^{(i-1)}|,\ |\rho^{(i)} - \rho^{(i-1)}|\big) < \text{tol}$.

Standard errors for gamma

The vcov of $\hat{\gamma}$ is the (2:end, 2:end) block of the inverse Hessian of the negative log-likelihood, evaluated at the optimum. The Hessian is computed by ForwardDiff. Index 1 corresponds to $\operatorname{atanh}(\hat{\rho})$, whose uncertainty is summarised separately via the delta-method formula $\mathrm{SE}(\hat{\rho}) = \sqrt{(1-\hat{\rho}^2)^2/n}$ shown in the display (consistent with the Fisher information for a stationary AR(1)).

Standard errors for beta

The OLS vcov on the doubly-transformed system, conditional on $(\hat{\rho}, \hat{\gamma})$.

Choices

- **L-BFGS over Nelder-Mead.** The log-likelihood is smooth and gradients are available cheaply via AD, making a quasi-Newton method much faster than a derivative-free method for this problem size.
- **Concentrated likelihood.** Profiling out β reduces the dimension of the optimisation from $k+1+p$ to $1+p$ (where k is the number of regressors and p the number of variance parameters), which substantially improves convergence.
- **ForwardDiff over FiniteDiff.** Automatic differentiation is exact to machine precision and avoids step-size tuning; the cost is proportional to the number of parameters, which is small here.

References

Oberhofer, W., & Kmenta, J. (1974). A general procedure for obtaining maximum likelihood estimates in generalized regression models. *Econometrica*, 42(3), 579–590.

Harvey, A. C. (1976). Estimating regression models with multiplicative heteroscedasticity. *Econometrica*, 44(3), 461–465.

Beach, C. M., & MacKinnon, J. G. (1978). A maximum likelihood procedure for regression with autocorrelated errors. *Econometrica*, 46(1), 51–58.

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11. Heteroskedasticity LM tests

Harvey test (`harvey_test`)

Tests $H_0 \colon \gamma = \mathbf{0}$ in the exponential variance model $\log \sigma_i^2 = \gamma_0 + \mathbf{z}_i^\top \gamma$.

Procedure. OLS residuals $\hat{u}_i \to$ auxiliary regression of $\log \hat{u}_i^2 - c$ on $\mathbf{Z}$ (same bias correction as in `exponential_fwls`) $\to$

$$\mathrm{LM} = \frac{\mathrm{ESS}}{\psi_1(1/2)} \sim \chi^2(p) \quad \text{under } H_0$$

where ESS is the explained sum of squares of the auxiliary regression, $\psi_1(1/2) = \pi^2/2 \approx 4.934$ is the trigamma function at 1/2 (i.e. the variance of $\log \chi^2(1)$), and $p = \dim(\gamma)$ (slope parameters only, excluding intercept).

Formula choice. The statistic is Harvey's (1976) exact score-test formula. A common approximation uses nR^2 instead, which is asymptotically equivalent but numerically different. HARE uses the exact formula, which matches `skedastic::harvey()` in R.

Glejser test (`glejser_test`)

Tests $H_0 \colon \gamma = \mathbf{0}$ in the quadratic variance model $\sigma_i = \gamma_0 + \mathbf{z}_i^\top \gamma$.

Procedure. OLS residuals $\hat{u}_i \to$ auxiliary regression of $|\hat{u}_i|$ on $\mathbf{Z} \to$

$$\text{LM} = \frac{\text{ESS}_{|\hat{u}|}}{\hat{\sigma}^2(1 - 2/\pi)} \sim \chi^2(p) \quad \text{under } H_0$$

where $\text{ESS}_{|\hat{u}|}$ is the explained sum of squares of the auxiliary regression, $\hat{\sigma}^2 = n^{-1} \sum \hat{u}_i^2$ is the OLS variance from the main regression, and $(1 - 2/\pi) = \text{Var}(|N(0,1)|)$ is the variance of the half-normal.

Formula choice. This is Glejser's (1969) exact score-test formula, matching `skedastic::glejser()` in R. The generic nR^2 approximation omits the $\hat{\sigma}^2(1 - 2/\pi)$ denominator and is asymptotically equivalent but numerically different.

Borrowed tests: Breusch-Pagan, White, Durbin-Watson, Breusch-Godfrey

Four additional diagnostic tests are provided as thin wrappers around `HypothesisTests.jl`, which supplies the underlying computations:

HARE function	Wraps	Null hypothesis
`breusch_pagan_test`	`WhiteTest(type=:linear)`	σ_i^2 constant (linear link)
`white_test`	`WhiteTest(type=:White)`	σ_i^2 constant (general)
`durbin_watson_test`	`DurbinWatsonTest`	no first-order serial correlation
`breusch_godfrey_test`	`BreuschGodfreyTest`	no serial correlation up to order p

Each wrapper runs OLS internally and accepts the same `(X, y; intercept=true)` and `(formula, data)` calling conventions as `harvey_test` and `glejser_test`, so users do not need to manage residuals manually.

Why snake_case names instead of the `HypothesisTests` CamelCase names?

`HypothesisTests.jl` already exports `BreuschPaganTest`, `WhiteTest`, `DurbinWatsonTest`, and `BreuschGodfreyTest` as `(X, e)` constructors, where `X` is the full design matrix (with intercept) and `e` is a pre-computed residual vector. HARE's convention is `(X, y)`, where `X` excludes the intercept and `y` is the raw response — the OLS step is handled internally. Both signatures are `(AbstractMatrix, AbstractVector)` positionally, so Julia cannot distinguish them by dispatch: defining HARE methods under the same CamelCase names would silently override the `HypothesisTests` constructors for those types, breaking any code that calls the original API directly (type piracy). Snake_case names (`breusch_pagan_test`, etc.) are a different namespace that coexists cleanly with `HypothesisTests`, and the distinction also makes the source of each name unambiguous to readers of the code.

Notes on γ_0 (intercept)

The FWLS estimator of γ_0 has residual variance $\text{Var}[\log \chi^2(1)] = \pi^2/2 \approx 4.93$, which is large. Monte Carlo at $N = 10{,}000$ (200 replications) shows:

Estimator	RMSE($\hat{\gamma}_0$)
`exponential_fwls` (FWLS)	0.021
`exponential_mle` (MLE)	0.013

The MLE is ~40% more efficient for γ_0. Stata's `hetregress` in MLE mode matches `exponential_mle`; in two-step mode it likely omits the bias correction (not documented), making its γ_0 biased in addition to being less efficient.

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12. Standard errors: common conventions

FWLS estimators (`exponential_fwls`, `exponential_ifwls`, `quadratic_fwls`, `quadratic_ifwls`, `power_fwls`, `power_ifwls`, Prais-Winsten, Sequential HARE) report **conditional** standard errors — conditional on the estimated nuisance parameters (ρ, γ). This matches the standard econometric textbook treatment (Greene, 2018; Judge et al., 1985) and produces valid inference asymptotically. Unconditional (sandwich-corrected) standard errors that account for two-step estimation uncertainty are not currently implemented.

MLE estimators (`exponential_mle`, `quadratic_mle`, `linear_mle`, `power_mle`, `groupwise_mle`, Joint HARE, Beach-MacKinnon) report standard errors from the **inverse observed information matrix**, computed by ForwardDiff at the optimum. For the heteroskedastic MLE estimators the full joint Hessian over (β, γ) is used, so the standard errors already account for the joint estimation of mean and variance parameters — no conditioning is needed.

The inference machinery supports both a t-distribution (default) and a normal distribution for p-values and confidence intervals. The t approximation is recommended for small samples.

Test design notes

Consistency testset: RNG and sample size

The `consistency (large n)` testset in `test/runtests_consistency.jl` uses `StableRNG(123)` from the StableRNGs.jl package with `N = 1_000_000` observations and `atol = 0.01`. These choices were made deliberately:

Why `StableRNG` instead of `Random.seed!`. Julia's default RNG implementation is not guaranteed to produce identical streams across Julia versions (e.g. 1.10 vs 1.12). `StableRNG` guarantees bit-identical draws on every platform and version, making the `atol` bounds platform-independent and eliminating spurious CI failures caused by RNG changes between releases.

Why `N = 1_000_000`. An earlier version used `N = 50_000` with `Random.seed!(2025)`. That seed produced an AR(1) draw whose residual mean was large enough to push the intercept estimate 0.021 above the true value, exceeding `atol = 0.02` and causing four failures (`two_step_prais_winsten`, `two_step_cochrane_orcutt`, `hildreth_lu`, `beach_mackinnon`). This was a sampling artifact, not a bias in the estimators. At `N = 1_000_000` the maximum error across all estimators and all parameters is below 0.007 for every seed tested (StableRNG(123), StableRNG(2025), StableRNG(42), StableRNG(999), StableRNG(31415), and three `Random.seed!` seeds), giving comfortable margin under `atol = 0.01`.

Why `atol = 0.01`. This is a tighter and rounder bound than the previous 0.02. The maximum observed error at `N = 1_000_000` is ~0.006, so the margin is still a factor of ~1.5. Users running the test suite with a different seed will see the same pass: the 1 % bound is robust at this sample size.

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